

Junyi Chen

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EDUCATION

-  **Shanghai Jiao Tong University** Sep 2024 – Present
(M.S. in Computer Science and Technology)
 - Advanced Networking Laboratory (NNE-Lab); Advisors: Prof. Fan Wu and Prof. Shengzhong Liu
 - GPA: 3.95/4.0
-  **Beihang University** Sep 2020 – Jun 2024
(B.S. in Computer Science and Technology)
 - Shen Yuan Honors College (honors program for the top 5% of students)
 - GPA: 3.92/4.0 (Rank: 1/50)

RESEARCH INTERESTS

- Efficient AI Algorithms
- LLM Post-Training and Inference
- Reinforcement Learning Systems and Machine Learning Systems

RESEARCH AND INTERNSHIP EXPERIENCE

-  **RED AI-Infra Training Framework Team, Xiaohongshu** Jul 2026 – Present
(ACE Talent Program · Research Intern)
 - Contribute to the development and maintenance of **Relax**, an open-source RL post-training framework for LLMs, and to performance analysis during Agentic RL rollouts; explore applications of inference acceleration techniques in post-training and participate in research on LLM post-training algorithms
 - Study inference acceleration across both MoE Prefill and Decode; based on stage-specific expert budgets and pairwise scalar projections, co-design the ExFold method and contribute to its system implementation
-  **LLM Systems and Toolchain Team, SenseTime Research** Dec 2023 – Jul 2026
(Research Intern)
 - Core contributor to LightLLM**, responsible for developing and maintaining decoding algorithms and model inference modules and for researching LLM inference acceleration
 - As first author, led the algorithm and system development of Pre³ and LightSpec: Pre³ uses a DPDA for low-overhead structured decoding, while LightSpec schedules speculative configurations online according to runtime Draft/Verify costs; integrated both into LightLLM's standard inference pipeline

HONORS AND AWARDS

ACL 2025 Outstanding Paper Award	Aug 2025	SenseTime Research "Rising Star" Award	Apr 2026
National Scholarship (Graduate)	Sep 2025	National Scholarship (Undergraduate)	2021 / 2023
Beijing Outstanding Graduate	2024	Beihang University Outstanding Graduate	2024

RESEARCH PROJECTS

- LightLLM** (Python) 
Distributed inference framework for high-throughput, low-latency LLM serving. As a core contributor, maintain decoding algorithms and model inference modules and develop inference acceleration techniques.
 - Pre³**: Moves prefix-conditioned edge analysis of LR(1) grammars to preprocessing and converts the transition graph into a DPDA, eliminating runtime path exploration through parallel deterministic transitions; after integration into LightLLM's standard decoding path, reduces TPOT by up to 40% and improves throughput by up to 36% over XGrammar
 - LightSpec**: Builds the first open-source dynamic speculative decoding system and formulates Draft/Verify configuration selection as closed-loop online scheduling; its Budget-then-Fill scheduler allocates verification budgets and selects Draft configurations from measured costs and effective progress, uniformly supporting MTP, autoregressive, and parallel-block drafters; improves request throughput by up to 69% over fixed configurations and 10.3% over the best static configuration
- Relax** (Ray Serve / Megatron-LM / SGLang) 
Omni-modal RL post-training engine built on Ray Serve, with Megatron-LM and SGLang as its training and inference backends; contribute to parts of the Rollout module's development and maintenance.
 - Performance analysis and exploration**: Participate in evaluating speculative decoding, KV-cache compression, and related techniques during Rollout, focusing on generation throughput, end-to-end overhead, and their evolution over training; explore adaptive trajectory allocation and generation-length control to improve trajectory-sampling efficiency

TokenFlow: LLM Text Streaming Serving System (Python / SGLang / PyTorch)

- **Core innovation:** Models bursty LLM serving as consumption-rate-constrained text streams; optimizes delivered tokens, TTFT, and stalls with a buffer-aware two-stage preemptive scheduler that balances progress using buffer occupancy, consumption rate, and switching overhead
- **System and results:** Co-designs write-through hierarchical KV caching, chunked write-back, and load-evict overlap to hide preemption I/O; improves effective throughput by up to 82.5% and cuts P99 TTFT by 80.2% without reducing total token throughput

C²KV: Trainable Compression and Position-Agnostic KV-Cache Reuse (PyTorch / HuggingFace)

- **Core innovation:** Proposes trainable KV compression and position-agnostic cache reuse; a lightweight C²Extractor encodes each document as compressible, composable KV modules, reducing non-prefix recomputation and long-context KV storage and transfer
- **Training and results:** Jointly trains learnable C²Tokens, separate QKV projections, and structured attention while freezing the base model; preserves semantics across reordered and multi-document composition and accelerates long-context inference by up to 17× at matched quality

SmartThinker: Progressive CoT Length Calibration (Verl)

- **Core innovation:** Models the inverted-U relation between reasoning length and accuracy; estimates a problem-specific optimal length from group-level Gaussian statistics over all and correct trajectories, compressing only overlong correct trajectories
- **Reward design and results:** Dynamically solves reward coefficients while preserving advantage signs without a tuned global compression strength; cuts reasoning length by up to 52.6% while improving accuracy, with up to 16.6% relative accuracy gain on AIME25

My contribution: Co-developed difficulty-adaptive length rewards and ran Math Reasoning RL training with Verl

ExFold: Unified Training-Free MoE Expert Folding (vLLM / Triton / CUDA)

- **Core innovation:** Unifies compute-bound Prefill and bandwidth-bound Decode as stage-specific output approximation; calibrates pairwise scalar projections on unlabeled data to fold removed-expert contributions into retained experts
- **System and results:** Implements a pluggable vLLM module and lightweight Triton/CUDA operators while preserving Fused-MoE; accelerates TTFT by up to 1.41× and TPOT by up to 2.46×, improves offline throughput by 20%, and retains about 99% average quality

My contribution: Unified MoE optimization across Prefill/Decode and co-designed the expert-folding method

PUBLICATIONS

- [1] **Pre³: Enabling Deterministic Pushdown Automata for Faster Structured LLM Generation** (*ACL 2025 Oral*, 🏆 **Outstanding Paper**)
Junyi Chen, Shihao Bai, Zaijun Wang, Siyu Wu, Chuheng Du, Hailong Yang, Ruihao Gong, Shengzhong Liu, Fan Wu, Guihai Chen
- [2] **TokenFlow: Responsive LLM Text Streaming Serving under Request Burst via Preemptive Scheduling** (*EuroSys 2026*)
Junyi Chen, Chuheng Du, Renyuan Liu, Shuochao Yao, Dingtian Yan, Jiang Liao, Shengzhong Liu, Fan Wu, Guihai Chen
- [3] **LightSpec: Resilient Speculative Decoding for Scalable LLM Serving via Dynamic Scheduling** (*KDD 2027, Under Review*)
Junyi Chen, Shihao Bai, Zaijun Wang, Huiyang Xu, Chenzhi Hu, Xintong Zhao, Zhou Ting, Hengsheng Lun, Ruihao Gong, Shengzhong Liu, Fan Wu, Guihai Chen
- [4] **C²KV: Compressed and Composable KV Cache Reuse for Efficient LLM Inference** (*KDD 2026*)
Chuheng Du, Junyi Chen, Hanlin Tang, Kan Liu, Tao Lan, Lin Qu, Chaoyue Niu, Shengzhong Liu, Guihai Chen, Fan Wu
- [5] **SmartThinker: Progressive Chain-of-Thought Length Calibration for Efficient Large Language Model Reasoning** (*ICML 2026*)
Chenzhi Hu, Qinzhe Hu, Yuhang Xu, Junyi Chen, Ruijie Wang, Shengzhong Liu, Jianxin Li, Fan Wu, Guihai Chen
- [6] **ExFold: Unified Expert Folding for Training-Free MoE Prefill-Decode Acceleration** (*AAAI 2027, Under Review*)
Juntong Wu, Yifei Liu, Junyi Chen, Siqi Fan, Chaoran Feng, Minghao Li, Zhangliujie, Weihang Chen, Li Yuan
- [7] **StructureBench: A Unified Benchmark Suite for Multi-Scenario Structured Generation Tasks with On-Device Models** (*IJCAI 2026*)
Xiaokun Xiong, Zhengjie Xu, Junyi Chen, Shihao Bai, Ruihao Gong, Xianglong Liu
- [8] **TinyFormer: Efficient Transformer Design and Deployment on Tiny Devices** (*IEEE TCAS-I 2026*)
Jianlei Yang, Jiacheng Liao, Fanding Lei, Meichen Liu, Lingkun Long, Junyi Chen, Han Wan, Bei Yu, Weisheng Zhao
- [9] **Maven: Fine-Grained Multimodal Alignment for Vehicle Sensing Under Spatiotemporal Mismatch** (*AAAI 2027, Under Review*)
Bowen He, Chenzhi Hu, Yatong Chen, Weiguo Wang, Yuan Wang, Yao Yang, Bohong Xiao, Junyi Chen, Shengzhong Liu, Chaoyue Niu, Fan Wu, Guihai Chen
- [10] **AdaQuant: Accelerating Low-Bit Quantized DNNs on Edge GPUs via Activation-Centric Intra-Operator Blending** (*SenSys 2027, Under Review*)
Qian Zan, Yatong Chen, Huiyang Xu, Wei Wang, Junyi Chen, Renyuan Liu, Shuochao Yao, Maohua Chen, Ting Zhou, Shengzhong Liu, Fan Wu, Guihai Chen

TEACHING AND ACADEMIC SERVICE

Operating Systems Lab (Teaching Assistant, Spring 2023): Designed course projects and the backend of Osome, an automated lab and grading platform serving nearly 600 students across four Beihang schools each semester

Academic Service: Reviewer for ACL, AAAI, KDD, and other international conferences